INCH POUND

MIL-PRF-38230D
w/AMENDMENT 1
1 November 2011
SUPERSEDING
MIL-PRF-38230D
26 February 2008

PERFORMANCE SPECIFICATION

TRANSMITTER, PRESSURE, OIL, VARIABLE RELUCTANCE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers oil pressure transmitters.

1.2 Classification. Transmitters will be of the following type, as specified (see 6.2):

TRU-66/A 100 percent (200 psi) overpressure

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. These lists do not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet the requirements specified in the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Oklahoma City Air Logistics Center/ENEP, 3001 Staff Drive, Suite 2AG68A, Tinker AFB, OK 73145-3036, or emailed to ocalc.enrs.sma@tinker.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https:assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25435	-	Transmitters, Pressure, Aircraft, Variable Reluctance Type, General Specification for
MIL-I-25437	-	Indicators, Aircraft, Pressure, Voltage Ratio Type, General Specification for
MIL-I-25438	-	Indicator, Pressure Type MO-2
MIL-DTL-27189	-	Indicator, Pressure, Oil
MIL-I-38133	-	Indicator, Pressure, Oil

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	-	Reliability Test Methods, Plans and Environments for Engineering, Development Qualification, and Production, Handbook for

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(ASQ documents may be obtained online at http://www.asq.org or from American Society for Quality, P.O. Box 3005, Milwaukee, WI 53201-3005 or 600 North Plankinton Avenue, Milwaukee, WI 53203, USA.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The transmitters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 General requirements. The requirements of MIL-PRF-25435 apply as requirements of this specification with the exceptions and additions specified herein.

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3.3 Workmanship. The transmitter, including all parts, shall be constructed in accordance with commonly accepted industrial workmanship standards.

3.4 Interface.

3.4.1 Pressure indicators. The transmitter shall sense the differential of two pressures, convert the differential pressure to an electrical signal, and provide the unamplified signal to an indicator conforming to MIL-I-25437, MIL-I-25438, MIL-PRF-27189, or MIL-I-38133. External wiring between the transmitter and indicator shall be as shown in figure 1.

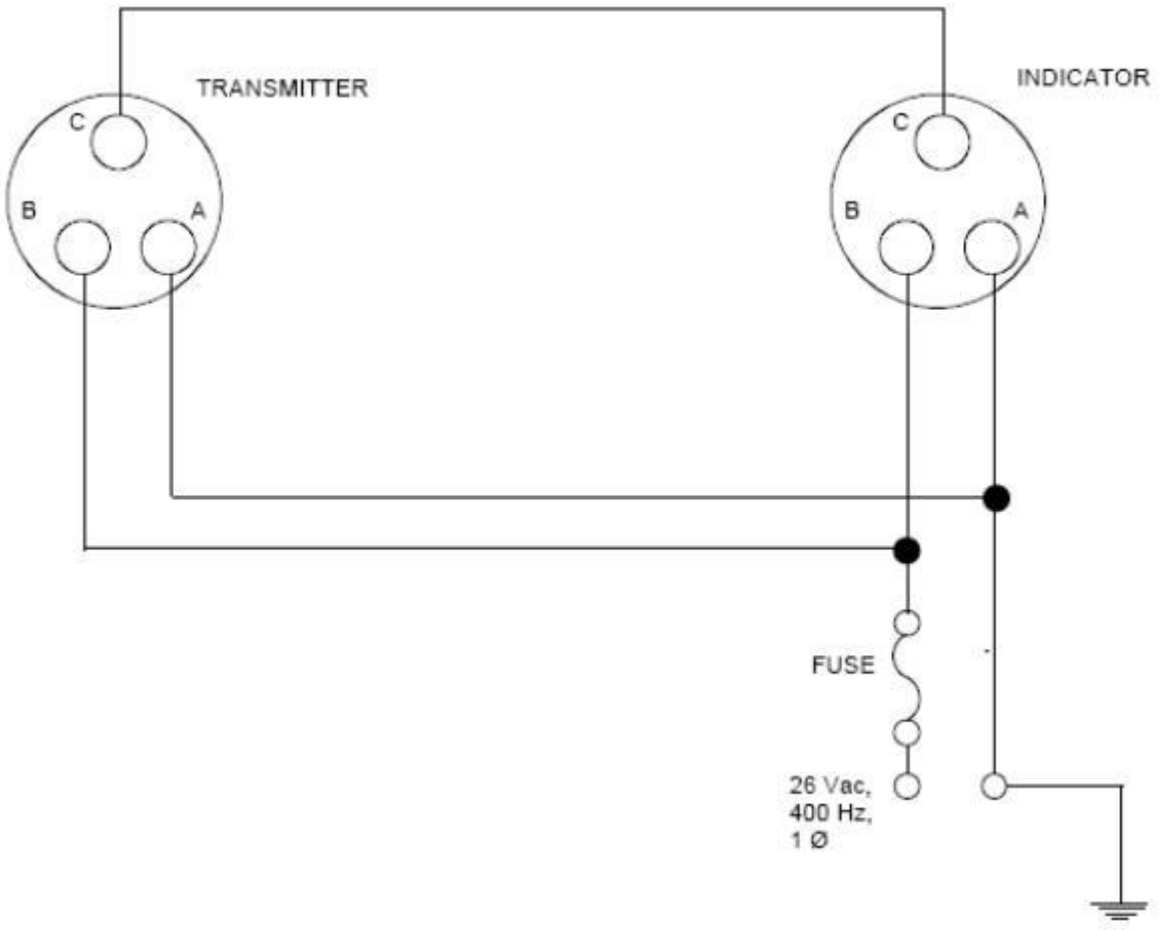


FIGURE 1. External wiring.

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3.5 Performance.

3.5.1 Transmitter error tolerances. The transmitter shall meet the scale and, friction error tolerances specified in tables I and II under the conditions noted for the tests herein and in MIL-PRF-25435. The transmitter shall operate within these tolerances when connected as shown in figure 1 to a standard indicating unit. The standard indicating unit shall be calibrated to a maximum readout error of ±0.25 percent using the standard ratio transformer circuit in figure 2.

Table I. Scale error.

Test points (psi)	Scale error tolerances (±psi)		
	Note 1	Note 2	Note 3
0	1.0	1.75	2.0
10	1.0	1.75	2.0
20	1.0	1.75	2.75
30	1.75	2.25	2.75
40	1.75	2.25	2.75
50	1.75	2.25	2.75
60	1.75	2.25	2.75
70	1.75	2.25	2.75
80	1.75	2.25	2.75
90	2.0	2.5	3.0
100	2.0	2.5	3.0
Note 1: Tolerance for the scale and friction error test at ambient temperature. Note 2: Tolerance after endurance (10,000 cycles), humidity, power, and overpressure tests. Note 3: Tolerance after vibration, low temperature, high temperature, endurance (60,000 cycles), temperature-altitude, and low temperature exposure tests. Also, the tolerance after the transmitter has been subjected to the 80-hour high temperature exposure test and is returned to ambient temperature for minimum of 4 hours.			

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Table II. Friction, position, and vibration error

Test	Test tolerance (± psi)
Friction error (Note 1)	
Ambient temperature	0.5
Low temperature	1.5
High temperature	0.5
Endurance (10,000 cycles)	0.5
Endurance (60,000 cycles)	0.5
Temperature-altitude	1.5
Low temperature exposure	1.5
High temperature exposure	0.5
Humidity	0.5
Power	0.5
Overpressure	0.5
Position error at 50 psi test point only	0.5
Vibration error (while vibrating)	
Pointer oscillation	0.5
Pointer variation	0.5
Note 1: Friction error will be tested at test points in table I.	

3.5.2 Overpressure. The TRU-66/A transmitter shall withstand a 100% (200 psi) overpressure for a maximum of 10 minutes without being damaged.

3.5.3 Wire length. The transmitter shall operate as specified through transmitter-to-indicator connections equal to or less than 300 feet using No. 20 AWG stranded copper wire.

3.5.4 Weight. The weight of the transmitter shall be 1.0 pound or less.

3.5.5 Reliability. The transmitter shall have a minimum mean time between failures (MTBF) of 2,000 hours.

3.5.6 Operating life span. The transmitter shall have a minimum operating life span of 3,000 hours.

3.6 Item identification. The transmitter shall be permanently and legibly marked with the following information (see 6.2).

a. Manufacturer’s name
b. Manufacturer’s CAGE code
c. Manufacturer’s part number
d. Manufacturer’s lot number
e. National stock number

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f. Transmitter type
g. Date of manufacture

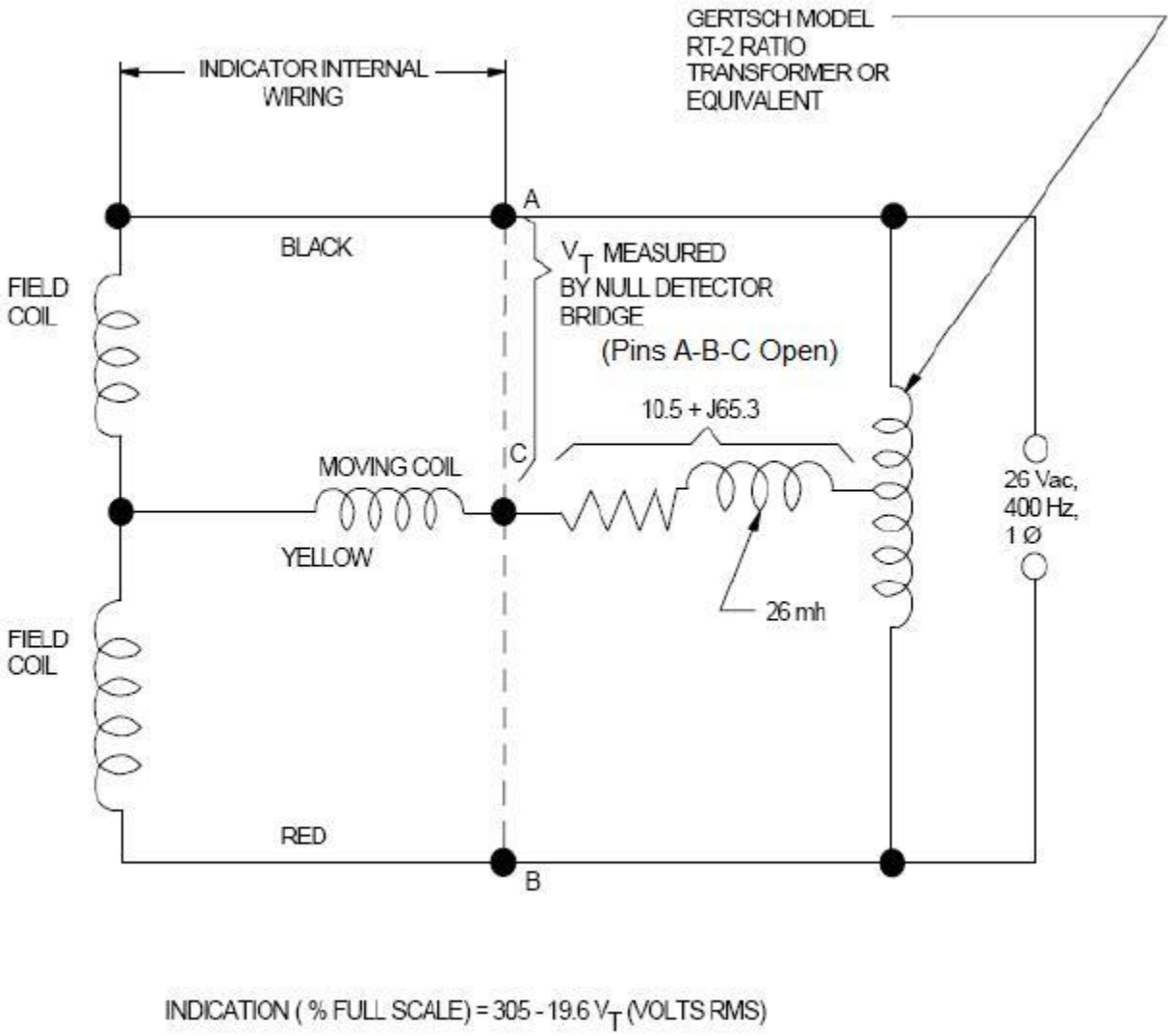


FIGURE 2. Test standard circuit.

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4. VERIFICATION

4.1 Inspection and test requirements. The inspection and testing of transmitters shall be as specified herein and in MIL-PRF-25435. This document shall govern in the event of duplication or conflict with MIL-PRF-25435. Tolerances shall be as stated herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection.
b. Conformance inspection.

4.3 Qualification inspection. Three transmitters shall be subjected to all tests herein and MIL-PRF-25435.except the reliability test and the fungus test. The reliability test is required for qualification and shall be performed on ten transmitters. One transmitter shall be subjected to the fungus test in MIL-PRF-25435.

4.4 Conformance inspection.

4.4.1. Individual tests. All transmitters shall be subjected to the individual tests of MIL-PRF-25435. The scale error and friction tests, performed alone and as part of other tests, shall be as herein.

4.5 Tests.

4.5.1 Scale and friction test. The transmitter shall be connected as shown in figure 1 to the standard indicating unit through 300 feet of No. 20 AWG stranded copper. The transmitter shall be subjected to the scale and friction error test of MIL-PRF-25435 at the test points in table 1 herein. Tolerances for the scale error and friction test, performed alone or as accept/reject criteria for other tests, shall be those in tables I and II.

4.5.1.1 Example. To verify that the transmitter meets the scale error tolerance after performing the overpressure test, refer to the Note 2 column and select a test point. If 20 psi is selected, the tolerance is ±1.75 psi. The standard indicating unit shall read 18.25 to 21.75 psi when performing the scale error test. The friction error allowed for this test is ±0.5 psi as shown in table II. Friction error is the difference in readings observed before and after tapping the transmitter while applying the test pressure of 20 psi.

4.5.2 Vibration. The transmitter shall be subjected to the vibration test in MIL-PRF-25435. The pointer oscillation and variation tolerances for the vibration error in table II shall be verified while conducting the vibration test.

4.5.3 Reliability. The reliability test is required for qualification unless waived in the acquisition document (see 6.2). See MIL-HDBK-781 for guidance.

4.5.3.1 Preparation. The transmitter shall be mounted directly to a vibration exciter without

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vibration isolators and shall be placed in a chamber for temperature cycling. The transmitter shall be connected to electrical power and the standard indicating unit as shown in figure 1. Pressure shall be applied to the transmitter as required for a standard indicating unit reading in the range of 40 to 60 psi.

4.5.3.2 Test. Rated power (see figure 1) and pressure shall be applied to the transmitter for 4 hours. The transmitter shall be subjected to random vibration for a minimum of 10 minutes during each hour when power is applied. The level of the acceleration power spectral density (APSD) shall be: 0.025 g^2/Hz at 15 Hz increasing at a rate of 6 dB/Octave to 0.3 g^2/Hz at 52 Hz; maintain 0.3 g^2/Hz from 52 to 852 Hz; decrease at a rate of -12 dB/Octave to 0.01 g^2/Hz at 2,000 Hz. The axis of vibration shall be longitudal. Temperature cycling shall be conducted as follows: the transmitter shall be brought to -65°F and maintained at that temperature for 4 hours; increase the transmitter temperature to 330°F at a minimum rate of 9°F per minute and maintain at that temperature for 4 hours; decrease the transmitter temperature to -65°F at a minimum rate of 9°F per minute and maintain at that temperature for 4 hours. Continue this sequence during the reliability test. The electrical power shall be turned off for 10 minutes after each 4 hours with power applied to the transmitter. Testing shall be halted and the transmitter shall be subjected to the scale error and friction test in MIL-PRF-25435 after each 250 hours of testing. The tolerances shall not exceed 150% of the ambient temperature tolerances in tables I and II. Transmitter malfunctions shall be recorded and repairs made. The transmitters shall not be recalibrated during the reliability test. The reliability test shall demonstrate a transmitter MTBF of 2,000 hours with a confidence level of 90% and a discrimination ratio of 1.5.

4.5.4 Operating life span. After satisfactory completion of the reliability test, and if specified in the acquisition document (see 6.2) an operating life span of 3,000 hours shall be demonstrated by continuing the reliability test. The transmitter shall be repaired when failures occur and testing continued until the transmitter has operated 3,000 hours.

4.5.5 Compatibility. The transmitter shall be connected to and tested with indicators specified in MIL-I-25437, MIL-I-25438, MIL-I-27189, and MIL-I-38133. The transmitter and indicators shall be connected as shown in figure 1. The scale error test in MIL-PRF-25435 shall be performed at ambient temperature. The tolerance at each test point shall be the sum of the tolerances in table I of this document and table I of MIL-I-25438 at ambient temperature.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The transmitters covered by this specification are intended for use with variable-reluctance-type indicators conforming to MIL-I-25437, MIL-I-25438, MIL-I-27189, and MIL-I-38133 in oil pressure indicating systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Item identification (see 3.6).
c. Type of transmitter required (see 1.2).
d. Data to be submitted with the qualification samples.
e. If a reliability test is waived (see 4.4.4).
f. If an operating life span test is required (see 4.4.5).
g. Packaging requirements (see 5.1).

6.3 Qualification. The attention of the contractors is called to the requirements with respect to products requiring qualification. Awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the Qualified Products List (QPL38230) whether or not such products have actually been listed by that date. In order that the manufacturers may be eligible to be awarded contracts or purchase orders for the products covered by this specification, they are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/ENEP, 3001 Staff Drive, Suite 2AG68A, Tinker AFB, OK 73145-3036.

6.4 Subject term (key word) listing.

TRU-66/A
Variable reluctance
0-100 PSI

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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